

Section 6 Lesson:

What are Algorithms?

Algorithm are a set of step-by-step instructions or rules designed to solve a specific problem or accomplish a particular task.

Algorithms are similar to that of a cooking recipe, or steps to complete a household chore

Laundry Algorithm Example:

If we think about doing laundry, we can break it into the following steps:

1. Collect Dirty Laundry
2. Place in Washing Machine
3. Add Detergent (And other cleaners/softeners)
4. Start washing machine
5. While (WashingMachine is Running) - Wait
6. Remove clothes from Washing Machine
7. For each item:
 - a. IF (sensitiveltem) - Hang dry
 - b. ELSE - Place in dryer
8. Add Dryer Sheets to dryer
9. Start Dryer
10. While (Dryer is Running) - Wait
11. Remove clothes from Dryer
12. Fold Dry Clothing
13. Put away folded clean clothes into proper drawers

This brings algorithms into simple easy to understand steps that we can see in our day-to-day lives.

Coding Algorithms:

First, it's good practice to define the algorithm's workflow in a common easy to understand language like the Laundry Algorithm above. This helps us to fully understand the steps that the algorithm will be performing.

Next, we will need to break down these instructions step-by-step into code to be executed.

Other Lesson and Activity Examples:

1. Pythagorean Theorem
2. Mathematical FOIL problems
3. Calculate Max (Integer)
4. Calculate Double (Double)
5. Calculate Factorial (Integer)